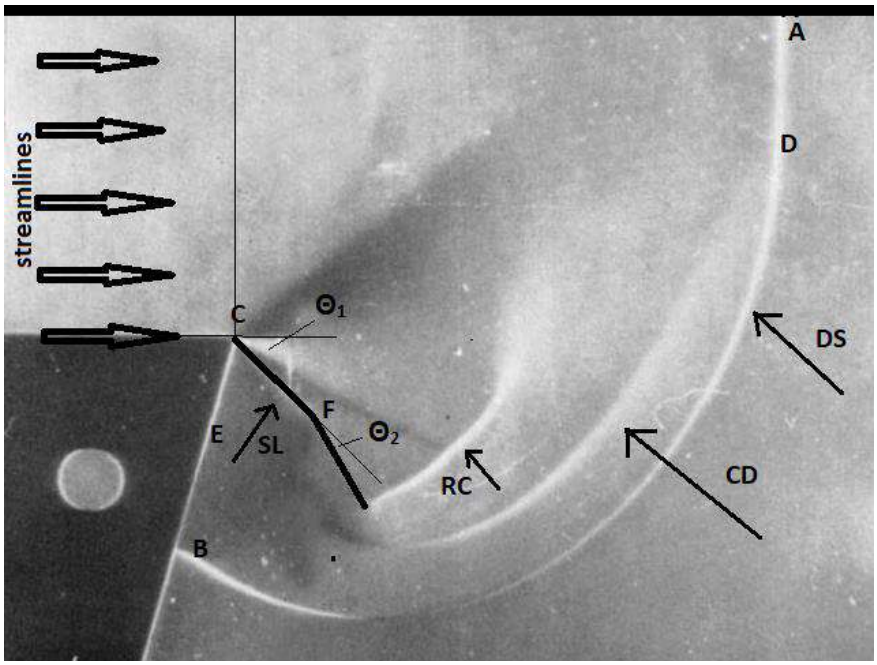
13.02.2023 - Gas Dynamics - MSAR - B. Favini

1. Diffracting Shock Wave (score 22/30)

A plane shock wave with Mach number equal 4 diffracts around a sharp corner (point C) made up of two plane walls with corner angle equal 105° (see schlieren photograph below). The gas in the undisturbed region ahead of the shock is air at ambient pressure p=1e5 pascal and temperature T=293 K. The ratio between the location of the head point A, distance measured from the abscissa of corner point C, and of the foot point B, distance measured along the inclined wall from corner point C, of the diffracted shock DS is equal to 2.5. The separation line (SL), starting from the corner C, shows a first deviation θ_1 of 44° and a second deviation θ_2 of 17° at point F.
Determine the flow state behind tha plane shock before the diffraction.
Determine the velocity of the foot, point B, of the diffracted shock.
Determine the strenght of the Prandtl-Meyer expansion issuing from the corner point C, justify its presence and explain why its head front ends in point D.
Determine the strenght of the Prandtl-Meyer expansion issuing at the point F.
Determine the strenght of the re-compression shock wave RC, justify its presence and its geometry.
Explain the presence of the contact discontinuity CD and why it extends from the point E, which moves along the inclined wall as the foot of the diffracted shock, up to point D.
Extend the streamlines along the entire flow field.
Determine the deviation of the separation line at corner C for the asymptotic flow field.



2. Boundary condition for pipe discharge (score 8/30)

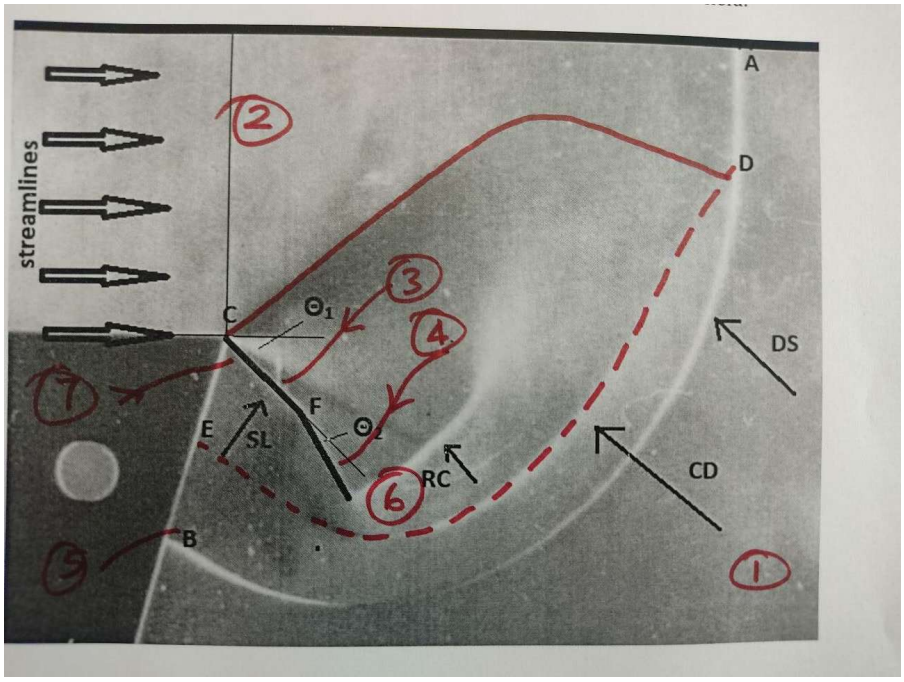
The exit section boundary conditions for an unsteady omentropic subsonic airflow along a pipe is:

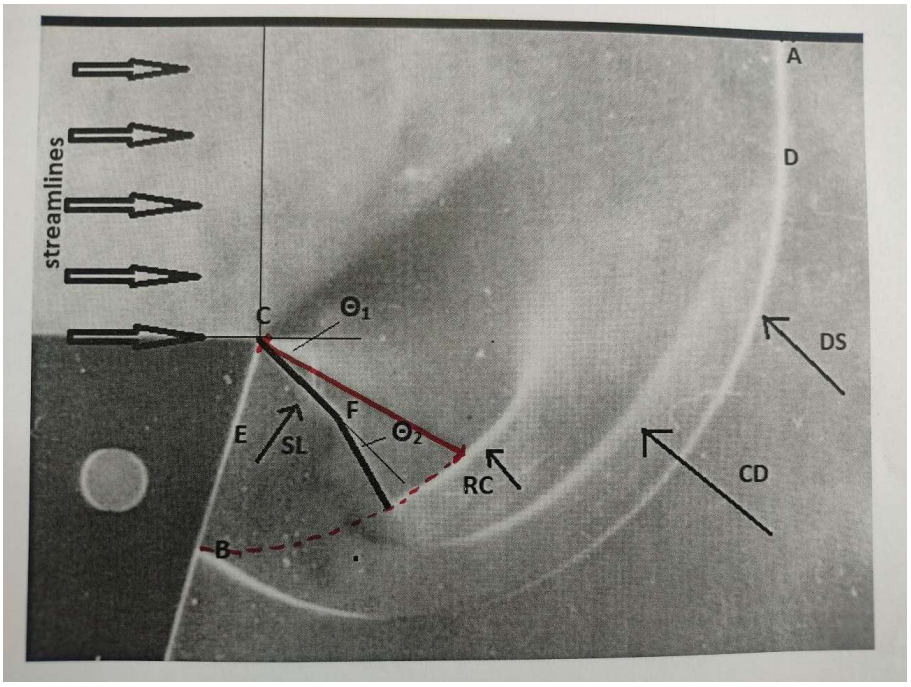
((γ+1) / 2) (u/a) ^ (4 / (γ+1)) + (a/u) ^ (2(γ-1) / (γ+1)) = K

with K=const.>1, u and a the velocity and the speed of sound, respectively, at the exit section.
Evaluate its effect on a acoustic simple wave propagating toward the exit section from inside the pipe.

Solutions 13.02.2023 - Gas Dynamics - MSAR - B. Favini

1. Diffracting Shock Wave (score 22/30)





Diffracting shock wave

state (1) $|\vec{V}_1| = 0$, $P_1 = 10^5$ Pa , $T_1 = 293$ K , $a_1 = 343{,}7$ m/s

state (2) behind the incoming plane shock $M = 4$

RH normal shock

$W_A = a_1 M = 1375$ m/s

$\frac{v_{behind}}{v_{behind}} = \frac{-W}{u_{behind} - W} = 4{,}5714$ $\quad u_2 = 1074$ m/s

$M_2 = \frac{u_2}{a_2} = 1{,}55$ $\quad a_2 = 691{,}4$ m/s

$P_2 = 18{,}5 \cdot 10^5$ Pa

state (3) behind the first P-M centered in point (C)

2D supersonic steady simple wave

$\nu(M_3) = \nu(M_2) + \theta_1 = 13{,}35° + 44° = 57{,}35°$

$M_3 = 3{,}43$

$P_3 = P_2 \frac{P_3/P_{03}}{P_2/P_{02}} = 1{,}059 \cdot 10^5$ Pa $\quad P_{02} = P_{03}$

$\{P_3 \perp P_1\}$

state (4) behind the second P-M centered in point (F)

as above

$\nu(M_4) = \nu(M_3) + \theta_2 = 57{,}35° + 17° = 74{,}35°$

$M_4 = 4{,}74$

$P_4 = P_3 \frac{P_4/P_{04}}{P_3/P_{03}} = 0{,}2 \cdot 10^5$ Pa $\quad P_{04} = P_{03}$

state (5) behind point (B)

shock velocity in (B) $W_B = W_A / 2{,}5 = 550$ m/s

RH normal shock

$M = \frac{W_B}{a_1} = 1{,}6$

$M_5 = \frac{u_5}{a_5} = 0{,}7$ $\quad P_5 = 2{,}82 \cdot 10^5$ Pa

$P_5 < P_2$ & $P_5 > P_3$

state (6) behind RC, which is unsteady and its velocity is almost equal to V_B, moreover it is not uniform

$M_3 = 3,43$ $a_3 = a_2 \frac{a_3/a_0}{a_2/a_0} = 458 \frac{m}{s}$ $|V_3| = 1571 \frac{m}{s}$

$M_3|_{RC} = \frac{|V_3| - W_B}{a_3} = 2,23$ $P_3 \leq 6 \cdot 10^5$ Pa

$M_4 = 4,74$ $a_4 = 358 \frac{m}{s}$ $|V_4| = 1697 \frac{m}{s}$

$M_4|_{RC} = \frac{|V_4| - W_B}{a_4} = 3,2$ $P_4 \leq 3,2 \cdot 10^5$ Pa

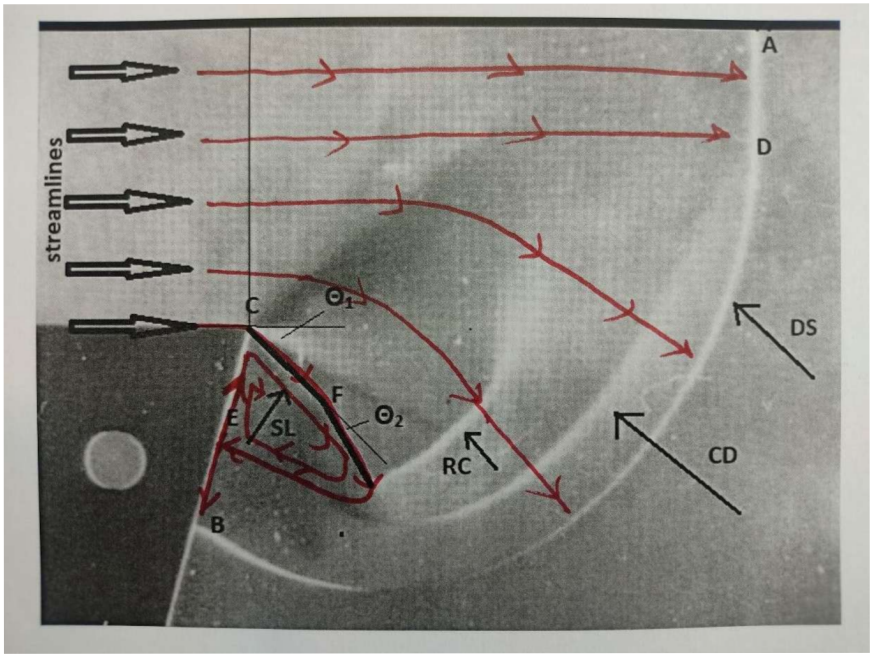
state (1) asymptotic state

$P_7 = P_1 \leq P_3 = P_{ambient} = 10^5$ Pa

$\theta_{separation\ line} = \theta_1 \leq 44°$

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Contact discontinuity extends from point (D), where the shock wave of diffracting, up to point (E) because the first particles of the state (2) have been affected by the plane shock, i.e. have the maximum level of entropy, but also some of them are affected also by RC, whereas the particles set in motion by the diffracted shock, that one in the region between the contact discontinuity and the diffracted shock, gains lesser entropy.


streamlines
A
D
C
Θ1
DS
F
E
SL
Θ2
RC
CD
B